

Filip Cano

Website: <https://filipcano.com>

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Research Interests

- Responsible artificial intelligence
- Runtime monitoring and enforcement of safety and fairness properties
- Accountability and explainability of autonomous agents
- Sequential decision-making

Education

PhD in Computer Science 03/2021–04/2025
Graz University of Technology (TUG)
Thesis: Towards Responsible AI: Advances in Safety, Accountability, and Fairness of Autonomous Systems
GPA: 1 (best mark in Austrian system). Passed with distinction.

MSc in Advanced Mathematics and Mathematical Engineering 09/2018–10/2019
Polytechnical University of Catalonia (UPC)
Thesis: An Introduction to Polytope Theory through Ehrhart's Theorem
GPA: 10/10 (1st of class)

BSc in Mathematics 09/2013–06/2018
Polytechnical University of Catalonia (UPC)
Thesis: Theoretical Study of Artificial Neural Networks
GPA: 8.8/10 (top 10%)

BSc in Physics Engineering 09/2013–06/2018
Polytechnical University of Catalonia (UPC)
Thesis: Theoretical Study of Artificial Neural Networks
GPA: 8.3/10 (top 25%)

Employment

Postdoctoral researcher 05/2025–Present
Institute of Science and Technology Austria (ISTA)
Group Thomas Henzinger.

University project assistant 03/2021–03/2025
Graz University of Technology (TUG)
Group Roderick Bloem and group Bettina Könighofer

Publications

CONFERENCE PAPERS

- [1] Filip Cano. “Explaining Decisions One Conversation at a Time: Opportunities and Risks of LLMs as Explainability Assistants”. In: *Proceedings of the International Conference on Agents and Artificial Intelligence (ICAART)*. 2026.
- [2] Filip Cano, Thomas A. Henzinger, and Konstantin Kueffner. “Energy Shields for Fairness”. In: *Proceedings of ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. ACM, 2026.
- [3] Filip Cano, Thomas A. Henzinger, Konstantin Kueffner, and N. Ege Saraç. “Monitoring Discounted Sum Properties”. In: *Proceedings of the International Conference on Concurrency Theory (CONCUR)*. 2026.
- [4] Amartya Hatua, Trung T. Nguyen, Filip Cano, and Andrew H. Sung. “Machine Unlearning using Forgetting Neural Networks”. In: *Proceedings of the International Conference on Agents and Artificial Intelligence (ICAART)*. 2026.
- [5] Filip Cano, Thomas A. Henzinger, Bettina Könighofer, Konstantin Kueffner, and Kaushik Mallik. “Fairness Shields: Safeguarding against Biased Decision Makers”. In: *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. AAAI Press, 2025.
- [6] Filip Cano, Thomas A. Henzinger, and Konstantin Kueffner. “Algorithmic Fairness: A Runtime Perspective (Invited Paper)”. In: *International Conference on Runtime Verification (RV)*. Vol. 16087. Lecture Notes in Computer Science. Springer, 2025, pp. 1–21.
- [7] Filip Cano, Thomas A. Henzinger, Bettina Könighofer, Konstantin Kueffner, and Kaushik Mallik. “Abstraction-Based Decision Making for Statistical Properties (Invited Paper)”. In: *International Conference on Formal Structures for Computation and Deduction (FSCD)*. 2024.
- [8] Samuel Judson, Matthee Elacqua, Filip Cano, Timos Antonopoulos, Bettina Könighofer, Scott J. Shapiro, and Ruzica Piskac. “‘Put the Car on the Stand’: SMT-based Oracles for Investigating Decisions”. In: *ACM Symposium on Computer Science and Law (CSLAW)*. 2024.
- [9] Samuel Judson, Matthee Elacqua, Filip Cano, Timos Antonopoulos, Bettina Könighofer, Scott J. Shapiro, and Ruzica Piskac. “soid: A Tool for Legal Accountability for Automated Decision Making”. In: *International Conference on Computer Aided Verification (CAV)*. 2024.
- [10] Saddek Bensalem, Panagiotis Katsaros, Dejan Nickovic, Brian Hsuan-Cheng Liao, Ricardo Ruiz Nolasco, Mohamed Abd El Salam Ahmed, Tewodros A. Beyene, Filip Cano, Antoine Delacourt, Hasan Esen, et al. “Continuous Engineering for Trustworthy Learning-Enabled Autonomous Systems”. In: *Bridging the Gap Between AI and Reality (AISoLA)*. 2023.
- [11] Katrine Bjørner, Samuel Judson, Filip Cano, Drew Goldman, Nick Shoemaker, Ruzica Piskac, and Bettina Könighofer. “Formal XAI via Syntax-Guided Synthesis”. In: *Bridging the Gap Between AI and Reality (AISoLA)*. 2023.

- [12] Filip Cano Córdoba, Samuel Judson, Timos Antonopoulos, Katrine Bjørner, Nicholas Shoemaker, Scott J Shapiro, Ruzica Piskac, and Bettina Könighofer. “Analyzing Intentional Behavior in Autonomous Agents under Uncertainty”. In: *International Joint Conference of Artificial Intelligence (IJCAI)*. 2023.
- [13] Filip Cano Córdoba, Alexander Palmisano, Martin Fränzle, Roderick Bloem, and Bettina Könighofer. “Safety Shielding under Delayed Observation”. In: *International Conference of Automatic Planning and Scheduling (ICAPS)*. 2023.
- [14] Martin Tappler, Filip Cano Córdoba, Bernhard K. Aichernig, and Bettina Könighofer. “Search-Based Testing of Reinforcement Learning”. In: *International Joint Conference of Artificial Intelligence (IJCAI)*. 2022.

TECHNICAL REPORTS

- [1] Jyotirmoy Deshmukh, Bettina Könighofer, Dejan Ničković, and Filip Cano. “Safety Assurance for Autonomous Mobility (Dagstuhl Seminar 24071)”. In: *Dagstuhl Reports* 14.2 (2024), pp. 95–119.
- [2] Bettina Könighofer, Joshua A. Kroll, Ruzica Piskac, Michael Veale, and Filip Cano Córdoba. “Accountable Software Systems (Dagstuhl Seminar 23411)”. In: *Dagstuhl Reports* 13.10 (2024), pp. 24–49.
- [3] Felipe Cano-Córdoba, Sanjay Sarma, and Brian Subirana. *Theory of intelligence with forgetting: Mathematical theorems explaining human universal forgetting using “forgetting neural networks”*. Tech. rep. Center for Brains, Minds and Machines (CBMM), Memo no. 71, 2017.

Teaching

MAIN INSTRUCTOR

- Model Checking. Institute of Science and Technology Austria. Spring 2026.

TEACHING ASSISTANT

- Interdisciplinary Science Communication. Institute of Science and Technology Austria. Main instructor: Giorgia Loretta. Fall 2026.
- Model Checking. Graz University of Technology. Main instructors: Roderick Bloem, Bettina Könighofer. Spring 2021–2024.

STUDENT THESIS CO-SUPERVISION (together with Bettina Könighofer)

- MSc. Fabio Rutter. Autonomous Single-Molecule Manipulation via Deep Reinforcement Learning. 2023.
- MSc. Stefan di Vora. Sending Mario back to school: an evaluation on how to speed up reinforcement learning using fuzzed traces. 2022.
- MSc. Johannes Haring. Adversarially robust transfer learning for image classification neural networks. 2022.
- BSc. David Broos. Safe Reinforcement Learning in Continuous State and Action Spaces. 2023.

- BSc. Philipp Rafael Rathmanner. Probabilistic Modelling of Vehicle Simulations in Autonomous Driving. 2023.
- BSc. Giacomo Pilotti. Continuous Action and Observation Space Environments for Reinforcement Learning. 2023.
- BSc. Mohammad Milad Tufan. Safe mapless navigation through Double Deep Q-Learning shielded with RSS. 2023.
- BSc. Lucy Burgstaller. Safe Reinforcement Learning in SafetyGym. 2022.

Honors and Grants

- Responsible AI Fellowship. 2026
Competitive funding from the Camp.org Foundation.
Grant for a 2-year postdoctoral researcher position at ISTA.
- Förderpreis des Forums Technik und Gesellschaft. 3rd Prize (1500€). 2026
Award for PhD thesis with most positive impact in society.
- National Graduation Award, 3rd Prize (2200€) 2024
Awarded by the government of Spain for top academic performance in university studies nationwide. Category Engineering and Architecture. Cohort graduating on 2017/18.
- Best Academic Record Award in the MSc program (2000€) . 2018/19.
Awarded by the Faculty of Math and Statistics at UPC.
- CFIS Mobility Grant for International Bachelor's Thesis (9000€). 2017/18.
Funded research stay at Massachusetts Institute of Technology, USA. Awarded by the Interdisciplinary Higher Education Centre (CFIS) of UPC.
- CFIS Study Grant 2013 – 2018.
Full tuition and housing scholarship for the whole BSc studies. Awarded based on admission ranking by CFIS.

Academic Service

EVENT ORGANIZATION

- Organization Committee Member: Young Scientists Symposium at ISTA, 2026.
- Volunteer program: IJCAI 2022 and 2023.

PEER REVIEWS

- Conference reviewer: AAAI (2025, 2026), AAMAS (2026), AISTATS (2026), EMNLP (2026), ICLR (2025, 2026), IJCAI (2024), ECAI (2025), FAccT (2026), NeurIPS (2026), VMCAI (2026).
- Journal reviewer: Science of Computer Programming (2024), Journal of Artificial Intelligence Research (2025).
- Conference sub-reviewer: AISoLA (2023, 2024), CAV (2025), FMCAD (2021 - 2023), FoSSaCS (2025), KR (2024), OVERLAY (2024), RV (2022), TACAS (2023, 2025), VMCAI (2024).

- Artifact evaluation: CAV (2023 - 2025), TACAS (2025, 2026), VMCAI (2024)

OTHER

- CFIS student mentoring program. 2016/17
Mentoring of 1st year BSc students of the double degree program of Maths and Physics Engineering.

Languages

Catalan. Native. Spanish. Native. English. Full proficiency. German. Basic.